

DIGVIJAY KEWALE

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Summary

GenAI and Automation Engineer with total 5+ years of professional experience, including 3+ years of hands-on expertise in Machine Learning, Generative AI, Automation, and Data Analytics. Proficient in Python, Scikit-Learn, TensorFlow, Power BI, and AI-driven automation solutions. Holds an M.Tech in Data Science and Machine Learning, with proven experience delivering innovative AI solutions through hackathons, and process automation initiatives that drive business impact and operational efficiency.

Technical Skills

Programming & Libraries: Python 3.0, NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn

Machine Learning & AI: Deep Learning, Neural Networks, NLP, TensorFlow, Keras, Transfer Learning

Databases & Analytics Tools: MySQL, Power BI, MS Excel, MongoDB, Firebase, Firestore

Deployment & Cloud: Flask, Streamlit, Docker, Git, GitHub Actions, Google Cloud

Professional Skills: Team Leadership, Collaboration, Problem-Solving

Experience

Capgemini Engineering

May 2023 – Present

Associate Engineer II

Bangalore, Karnataka

- Developed AI and GenAI applications using LLMs, computer vision like OCR (Paddle, Rapid), Python and automation frameworks to solve engineering and document-processing use cases.
- Built intelligent applications for PDF information extraction, workflow diagram generation, drawing review, engineering calculations, model updates and mid-surface extraction from 3D CAD models.
- Designed end-to-end AI automation pipelines integrating machine learning models, APIs and engineering software to streamline repetitive engineering processes and reduce manual intervention.
- Worked closely with engineering and technical stakeholders to identify AI automation opportunities, convert business requirements into scalable solutions, and support deployment of AI-driven applications.
- Delivered AI/GenAI training and knowledge-sharing sessions for cross-functional teams, covering AI concepts, tools, automation approaches and practical engineering use cases.

Sutra Systems India Pvt. Ltd.

Nov 2021 – May 2023

Design Engineer

Pune, Maharashtra

- Project: Galley, Palmet, Stowage — handled large assemblies and systems (electrical, water piping, air routing, cooling) and GA drawings for Airbus A330, A320, A380 and Boeing. Created 3D modeling and drafting of sheet metal and milling parts.
- Created drafting and Model Based Definitions (MBD) using PTC Creo 4.0 and operated Wind-chill PLM Software. Responsible for Engineering Changes and documentation.

Education

M.Tech in Data Science and Machine Learning

Jan 2024 – April 2026

PES University, Bangalore, Karnataka

CGPA: 9.26

B.E. in Mechanical Engineering

Aug 2012 – Sep 2015

G.H. Raisoni Academy of Engineering and Technology, Nagpur (RTMNU)

62.8%

Projects

Knowledge Grounded Conversational Agent using LLM and RAG | M.Tech Project, PES University

- Built an end-to-end **Multimodal RAG pipeline** combining FAISS-based dense retrieval, BGE embeddings, cross-encoder reranking, and **Gemini 2.0 Flash** for grounded response generation — achieving **41% hallucination reduction** vs. vanilla LLM and **91% top-3 retrieval relevance** after reranking.
- Extended text-only RAG to **multimodal** by integrating **CLIP vision-language embeddings** for image retrieval, enabling the agent to answer queries grounded in both documents and figures simultaneously.
- Deployed a **Streamlit web app** with Text-to-Speech support; modular architecture scaled seamlessly from text RAG to multimodal RAG with human evaluation scores of **4.7/5 factual grounding** and **4.8/5 fluency**.

Customer360: Turning Data into Insights using Recommendation System | M.Tech Project, PES University

- Developed a scalable recommendation system combining **K-Means clustering** for user segmentation and **SVD-based collaborative filtering**, delivering context-aware recommendations.
- Hybrid K-Means + SVD pipeline showed best performance; proposed **FAISS** integration for low-latency, large-scale retrieval.
- Planned enhancements include **BERT embeddings** for semantic understanding, user feedback mechanisms, and multilingual support.

Hackathons

Sahayak AI | Google Cloud Agentic AI Day Hackathon powered by Hack2Skill

- * Developed an **AI-powered teaching assistant** to support multi-grade classrooms in India, reducing teacher workload and enabling **personalized student learning**.
- * Designed scalable architecture on **Google Cloud**: Firebase Authentication, Firestore Database, and Cloud Functions.
- * Integrated **Gemini API** and **Dialogflow ES** for real-time insights and conversational support.
- * Built **React dashboard frontend** with secure data access for teachers.
- * Automated student performance tracking, enabling **40% faster intervention** for struggling learners.

Predicting Employee Attrition for a Fast-Growing Company

- * Developed a machine learning model to proactively predict employee attrition based on various employee attributes, helping HR teams take preventive measures to retain talent.
- * Built and evaluated multiple models using Accuracy as the primary metric. **RandomForestClassifier** performed best among all evaluated models.

Digvijay Kewale
Date: _____